CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech.)(Backlog)

University Examination April-May 2011

EE 201, Control Engineering

Date: 19/05/2011, Thursday **Time: 10:00 a.m. to 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Write both the sections in separate answer sheets.

2. Clearly mention the assumptions if any.

SECTION-I

			Marks
Q.1	(a)	Explain the difference between open loop and closed loop (feedback) control system by giving proper example of each system.	6
	(b)	Find the transfer function for the network shown in fig. 1.	6

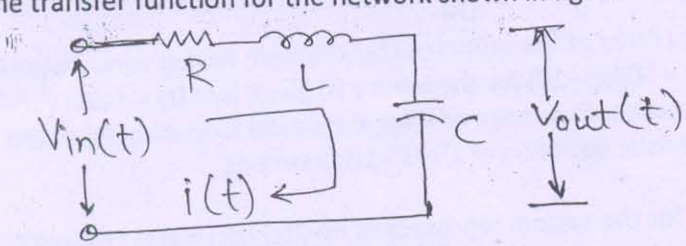

fig.1

Q.2	(a)	Find the transfer function of a system shown in fig.2 using block diagram reduction technique.	8

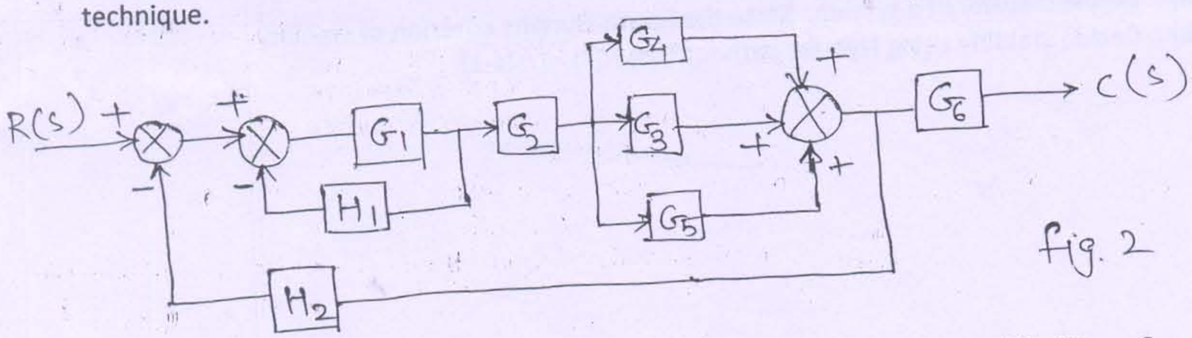

fig. 2

	(b)	Draw the signal flow graph representation for the system shown in fig. 2 and find its transfer function using Mason's gain formula.	8

OR

Q.2	(a)	Find the transfer function of the system represented by Signal flow graph in fig. 3 using Mason's gain formula.	8

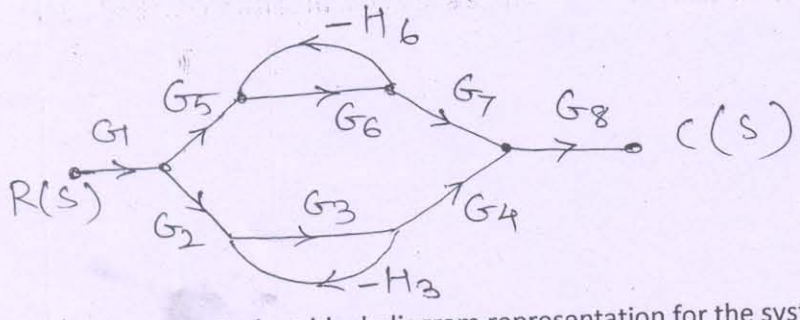

fig. 3

	(b)	Draw equivalent block diagram representation for the system shown by fig. 3 and	8

[P.T.O.]

find the transfer function using block diagram reduction technique.

Q.3 (a) Define gain margin, phase margin. Explain the system stability concept based on gain margin and phase margin. 5

(b) Write a short note on lead-lag compensator. 2

OR

Q.3 Draw the bode plot for the system represented by transfer function G(s) H(s) = 10/s(s+1) (s+10). 7

SECTION-II

Q.4 (a) Write a short note on PID controller. 4

(b) Derive the unit step response of a second order system. 8

Q.5 (a) Define steady state error. How to determine type of a system? 4

(b) Plot the root locus diagram for G(s)H(s)=k/s(s+3)(s+6). 8

OR

Q.5 (a) Estimate the steady state error of the unity feedback system having forward path transfer function as G(s) = 50/s(s+10) for the input r (t) given by r (t) = $1+2t+t^2$. 4

(b) Using Routh- Hurwitz criterion, find range of K for the closed loop stability of the system given by characteristic equation of $s^4+7s^3+10s^2+ks+k=0$. 8

Q.6 (a) Find number of root loci for the system represented by G(s) H(s) = k(s+1)/s(s+2)(s+3). 4

(b) Decide stability using Nyquist path for G(s) H(s) =1/s(s+1). 7

OR

Q.6 (a) Define stability of a system. State the Routh-Hurwitz criterion of stability. 4

(b) Decide stability using Nyquist path for G(s) H(s) =1/s(s-1). 7